

ALGORITHMS | LEVEL 2 | LAB 2

Big-O Notation Part II

Example #1:

Compute S.E (step execution) and big O for the following code:

```
int s = 0;           → 1
int i = 1;           → 1
while(i <= n)        → n-1+1+1 = n+1
{
    s = s + i;        → n
    i++;              → n
}
return s;            → 1
```

$$\text{S.E} = 1 + 1 + n + 1 + n + n + 1 = 3n + 4 \in O(n)$$

Example #2 : for loop increment by more than 1

```
for(int i=1; i < n; i = i+4) →  $\frac{\text{End} - \text{Start}}{4} + 1$ 
{
    S.O.P();                →  $\frac{\text{End} - \text{Start}}{4} + 1$ 
}
```

Exercise #1: for loop increment by 3

```
int sum=0;           → 1
for(int i=0; i < n; i = i+3) →  $\frac{n}{3} + 1$ 
{
    S.O.P();          →  $\frac{n}{3}$ 
    sum+=i;           →  $\frac{n}{3}$ 
}
```

$$\text{S.E} = 1 + \frac{n}{3} + 1 + \frac{n}{3} + \frac{n}{3} = 3\frac{n}{3} + 2 = n + 2 \in O(n)$$

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Example #3 : Nested for & Inner Loop j it depends on Outer Loop i

```

int x = 1;           → 1
for(int i = 1; i <= n; i++) → n-1+1+1 = n+1
{
    sum = 1;         → n
    for(int j = 1; j < i ; j++) →  $\frac{n(n+1)}{2} + 1$ 
    {
        x = i + j;   →  $\frac{n(n+1)}{2}$ 
        y = x + 3;   →  $\frac{n(n+1)}{2}$ 
    }
}

```

$$S.E = 1 + n + 1 + n + \frac{n(n+1)}{2} + 1 + \frac{n(n+1)}{2} + \frac{n(n+1)}{2}$$

$$= \frac{n^2+n}{2} + \frac{n^2+n}{2} + \frac{n^2+n}{2} + 2n + 3$$

$$= \frac{1}{2} (3n^2 + 3n) + 2n + 3$$

$$Big O \in O(n^2)$$

Example #4: For Loop (Special Case): loop increment by i=i*2 or i=i/2

```

for (int i=1; i<=n ; i=i*2) →  $\log_2(n) + 1$ 
{
    sum=sum+i; →  $\log_n$ 
}

```

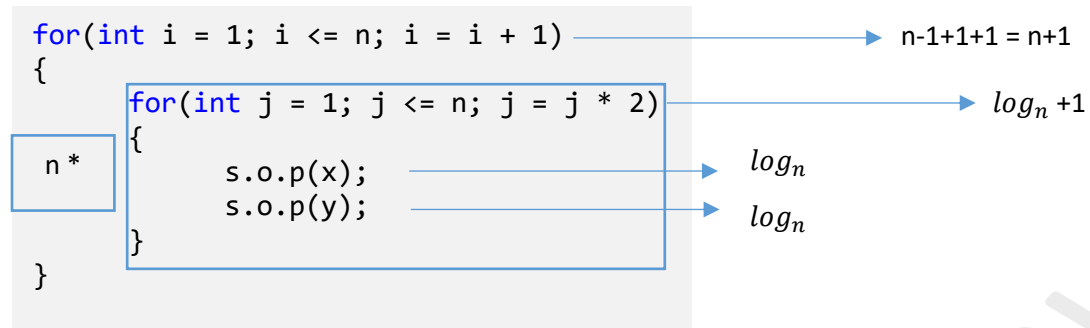
$$S.E = \log_n + 1 + \log_n$$

$$= 2 \log_n + 1$$

$$= O(\log_n)$$

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Exercise #2:



$$\begin{aligned} \text{S.E} &= n + 1 + n \log_n + n + n \log_n + n \log_n \\ &= 3 n \log_n + 2n + 1 \in O(n \log_n) \end{aligned}$$